Sagar Kriplani

Mechanical Engineer | Research & Development

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Mechanical engineer with an M.Sc. in Scientific Instrumentation, focused on thermal investigation, experimental setups and precision equipment development. Experience with vacuum polymer-heating research, mechanical design and engineering analysis.

EXPERIENCE

NeSt GmbH | Mechanical Design Engineer · Konstrukteur

09/2025 - 09/2026 | Herbrechtingen, Germany

Developed machine enclosures, precision assemblies, sheet-metal details and manufacturing drawings using Autodesk Inventor, with DFM/DFA and production coordination. Project figures reported by me: 8+ machine platforms, approximately 200 drawings per month, component cost reductions up to 70% and weight reduction of 60%. These figures are not independently established by the employer reference.

Brooks Automation | Master's Thesis Research & Mechanical Design

07/2023 - 06/2024 | Steißlingen, Germany

07/2023 - 01/2024: Research & Development (Master's Thesis) — compared radiant heaters and LEDs for polymer heating in vacuum chambers, using thermal models and experimental setups. 02/2024 - 06/2024: Mechanical Design Engineer (Working Student) — designed precision components and assemblies for semiconductor automation, developed CAD models and manufacturing drawings, and collaborated with R&D, electrical engineering and production.

Brooks Automation | Engineering Intern · Mechanical Design

01/2023 - 06/2023 | Steißlingen, Germany

Analysed the Process Air Unit, compared heater concepts and developed mechanical design and assembly documentation for semiconductor automation equipment. The issued internship reference reports 20% cost reduction; the project report separately records approximately 21%.

Ravi Brass Industries | Design & Development Engineer

08/2020 - 09/2021 | Jamnagar, India

Designed mechanical components and assemblies, prepared manufacturing drawings and coordinated with production on manufacturability and quality.

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PROJECTS & QUALIFICATIONS

Polymer heating · Master's thesis

Compared radiant heaters and LEDs in vacuum chambers. LEDs improved uniformity in the tested setup; radiant heaters performed better overall. Company research: 07/2023-01/2024; thesis submitted 02/2025.

PAU redesign · Brooks internship

Thermodynamic calculations, component selection and mechanical design. Project report: approximately 500 to 50 mbar pressure drop and 48 to 18 kW heater design capacity; not measured operating energy savings.

Academic CAD & simulation

Cylindrical handling mechanism in Inventor; ANSYS training exercises; MATLAB finite-difference frost-growth model in a four-student team. Academic results are not production validation.

EDUCATION

M.Sc. Scientific Instrumentation | 10/2021 - 03/2025

Ernst-Abbe-Hochschule Jena · Grade 1.8 · Thesis 1.3 · 120 ECTS · ECTS B

B.E. Mechanical Engineering | 2017 - 2020

A. D. Patel Institute of Technology / GTU · CGPA 8.14/10 · First Class with Distinction

Diploma in Mechanical Engineering | 2014 - 2017

Government Polytechnic Jamnagar / GTU · CGPA 9.50/10

TECHNICAL SKILLS

Autodesk Inventor | Siemens NX | PTC Creo | ANSYS | MATLAB | DFM/DFA | BOM | SAGE ERP | Agile PLM

TRAINING

ANSYS Basic to Professional (2020); Siemens Centre of Excellence: NX, NC, Drafting, Robotics (2016-2017); LinkedIn Learning: Siemens NX, Engineering Drawings, Semiconductor Manufacturing (2026).

LANGUAGES

German: professional working; English: full professional; Hindi, Gujarati, Sindhi: native/bilingual. Self-reported; German certificate: B1-plus coursework.

Redacted certificates, references and transcripts: sagar-kriplani.github.io/en/evidence/